

Quantum Mekaniği II

$$B = \frac{1}{c} \int \mathbf{J}(\mathbf{x}') \frac{(\mathbf{x} - \mathbf{x}')}{|\mathbf{x} - \mathbf{x}'|^3} d^3x' \Rightarrow \frac{1}{c} \nabla \times \int \mathbf{J}(\mathbf{x}') \frac{1}{|\mathbf{x} - \mathbf{x}'|} d^3x' \Rightarrow B = \nabla \times \underbrace{\frac{1}{c} \int \mathbf{J}(\mathbf{x}') \frac{1}{|\mathbf{x} - \mathbf{x}'|} d^3x'}_A \Rightarrow B = \nabla \times A$$

$$B' = \nabla \times A' \quad A' = A - \nabla f \quad B' = \nabla \times A' \Rightarrow B = \nabla \times A$$

$$E' = -\nabla \phi' - \frac{1}{c} \frac{\partial A'}{\partial t} \quad \phi' = \phi + \frac{1}{c} \frac{\partial f}{\partial t} \Rightarrow \nabla \times A' = \nabla \times (A - \nabla f) \quad \text{1)} E' = -\nabla \phi' - \frac{dA'}{dt} \quad \phi' = \phi + \frac{df}{dt}$$

$$\Rightarrow \nabla \times A' = \nabla \times A - \nabla (\nabla f) = B$$

$$\text{1)} \nabla \times A' = B' \quad \nabla \times A = B$$

$$E' = -\nabla(\phi + \frac{df}{dt}) - \frac{d}{dt}(A - \nabla f)$$

$$\Rightarrow E' = -\nabla \phi - \frac{d\nabla f}{dt} - \frac{dA}{dt} + \frac{d\nabla f}{dt}$$

$$E' = -\nabla \phi - \frac{dA}{dt} = E$$

Maxwell EQ

$$\nabla \cdot E = \rho/\epsilon_0$$

$$\nabla \times E = -\frac{dB}{dt}$$

$$\nabla \cdot B = 0$$

$$\nabla \times B = \frac{4\pi}{c} J + \frac{1}{c} \frac{dE}{dt}$$

$$\text{1)} \nabla \cdot E = \rho/\epsilon_0 \Rightarrow \nabla \cdot (-\nabla \phi - \frac{1}{c} \frac{dA}{dt}) = \rho/\epsilon_0$$

$$\Rightarrow -\nabla^2 \phi - \frac{1}{c} \frac{d}{dt} \nabla A = \rho/\epsilon_0$$

$$\nabla \cdot A = 0 \Rightarrow \text{coulomb gauge} \Rightarrow -\nabla^2 \phi = \rho/\epsilon_0 = \text{Poisson Denklemi}$$

$$\text{2)} \nabla \times B - \frac{1}{c} \frac{dE}{dt} = \frac{4\pi}{c} J \Rightarrow$$

$$\Rightarrow \nabla \times B - \frac{1}{c} \frac{d}{dt} (-\nabla \phi - \frac{dA}{dt}) = \frac{4\pi}{c} J \Rightarrow \nabla \times B + \frac{1}{c} \frac{d}{dt} \nabla \phi + \frac{1}{c^2} \frac{d^2 A}{dt^2} = \frac{4\pi}{c} J$$

$$\nabla \times B = \nabla \times (\nabla \times A) \Rightarrow \nabla(\nabla A) - \nabla^2 A$$

$$b(\partial c) - (ab)c$$

$$\Rightarrow \nabla(\nabla A) - \nabla^2 A + \frac{1}{c} \frac{d}{dt} \nabla \phi + \frac{1}{c^2} \frac{d^2 A}{dt^2} = \frac{4\pi}{c} J$$

$$\Rightarrow \nabla(\nabla A + \frac{1}{c} \frac{d\phi}{dt}) - \nabla^2 A + \frac{1}{c^2} \frac{d^2 A}{dt^2} = \frac{4\pi}{c} J$$

$$\nabla A + \frac{1}{c} \frac{d\phi}{dt} = 0 \Rightarrow \text{Lorentz gauge}$$

$$\Rightarrow \nabla A = -\frac{1}{c} \frac{d\phi}{dt} \quad \text{Lorentz gauge}$$

$$\nabla \cdot A = -\frac{1}{c} \frac{d\phi}{dt} \Rightarrow$$

$$\Rightarrow \text{Lorentz gauge} \quad \nabla \cdot E = 4\pi \rho \quad \text{uygusok}$$

$$\Rightarrow \nabla \cdot (-\nabla \phi - \frac{dA}{dt}) = 4\pi \rho \Rightarrow -\nabla^2 \phi - \frac{d}{dt} \nabla A = 4\pi \rho \Rightarrow -\nabla^2 \phi - \frac{d}{dt} (-\frac{1}{c} \frac{d\phi}{dt}) = 4\pi \rho$$

$$-\nabla^2 \phi + \frac{1}{c^2} \frac{d^2 \phi}{dt^2} = 4\pi \rho \Rightarrow \nabla^2 \phi - \frac{1}{c^2} \frac{d^2 \phi}{dt^2} = -4\pi \rho \left(\frac{1}{c} \right) = -\frac{4\pi}{c} J$$

$$\square_\mu^2 A_\mu = -\frac{4\pi}{c} J_\mu$$

De Lambert operatörü

$$\text{Zaman Bağı Sch} \Rightarrow i\hbar \frac{\partial \Psi}{\partial t} = -\frac{\hbar^2}{2\mu} \frac{\partial^2 \Psi}{\partial x^2} = E\Psi$$

Denk

$$\Rightarrow i\hbar \frac{\partial \Psi}{\partial t} = E\Psi$$

$$\Rightarrow \frac{\partial \Psi}{\partial t} = \frac{E}{i\hbar} \Psi \Rightarrow$$

$$\frac{\partial \Psi}{\partial t} = \frac{E}{i\hbar} \Psi \Rightarrow \ln \Psi = \frac{E}{i\hbar} t \Rightarrow \ln \Psi = \frac{E}{i\hbar} t + \ln C$$

$$\Rightarrow \left(\frac{\ln \Psi}{\ln C} = \frac{E}{i\hbar} t \right) \Rightarrow \Psi = C e^{-iEt/\hbar}$$

$$\Psi = e^{-iEt/\hbar}$$

$$H = \frac{p^2}{2\mu} \Rightarrow H = \frac{p^2}{2\mu} \Rightarrow H = \frac{1}{2\mu} (p^2 + \frac{e}{c} A) - e\phi$$

$$\frac{1}{2\mu} \left(\frac{\hbar}{i} \nabla + \frac{e}{c} A \right) \left(\frac{\hbar}{i} \nabla + \frac{e}{c} A \right) \Psi - e\phi \Psi = i\hbar \frac{\partial \Psi}{\partial t}$$

$$\Rightarrow A' = A - \nabla f \Rightarrow A = A' + \nabla f$$

$$\Rightarrow \frac{1}{2\mu} \left(\frac{\hbar}{i} \nabla + \frac{e}{c} (A' + \nabla f) \right) \left(\frac{\hbar}{i} \nabla + \frac{e}{c} (A' + \nabla f) \right) \Psi - e\phi \Psi$$

$$\frac{1}{2\mu} \left(\frac{\hbar}{i} \nabla + \frac{e}{c} A \right) \left(\frac{\hbar}{i} \nabla + \frac{e}{c} A \right) \Psi - e\phi \Psi = i\hbar \frac{d\Psi}{dt}$$

$$e^{i\lambda} \Psi \quad \Psi' \Rightarrow e^{i\lambda} \Psi' = \Psi$$

$$\Rightarrow A' = A - \nabla f \Rightarrow A = A' + \nabla f$$

$$\Rightarrow \frac{1}{2\mu} \left(\frac{\hbar}{i} \nabla + \frac{e}{c} (A' + \nabla f) \right) \left(\frac{\hbar}{i} \nabla + \frac{e}{c} (A' + \nabla f) \right) \Psi' - e\phi \Psi'$$

$$\frac{1}{2\mu} \left(\frac{\hbar}{i} \nabla + \frac{e}{c} (A' + \nabla f) \right) \underbrace{\left(\frac{\hbar}{i} \nabla + \frac{e}{c} A' \right)}_A e^{i\lambda} \Psi - e\phi e^{i\lambda} \Psi$$

$$\Rightarrow \# = \left(\frac{\hbar}{i} \nabla + \frac{e}{c} A \right) e^{i\lambda} \Psi = \frac{\hbar}{i} \nabla e^{i\lambda} \Psi + \frac{\hbar}{i} e^{i\lambda} \nabla \Psi + \frac{e}{c} A e^{i\lambda} \Psi =$$

$$e^{i\lambda} \left(\frac{\hbar}{i} \nabla \Psi + \frac{\hbar}{i} \nabla \lambda \Psi + \frac{e}{c} A \Psi \right) = \left(\frac{\hbar}{i} \nabla \Psi + \frac{e}{c} (A \Psi + \hbar \nabla \lambda \Psi) \right)$$

$$e^{i\lambda} \frac{1}{2\mu} \left(\frac{\hbar}{i} \nabla + \frac{e}{c} A \right) \left(\frac{\hbar}{i} \nabla + \frac{e}{c} (A + \hbar \nabla \lambda) \right) \Psi = i\hbar \frac{d}{dt} e^{i\lambda} \Psi$$

$$\Rightarrow i\hbar \left(\lambda \frac{d}{dt} e^{i\lambda} \Psi + e^{i\lambda} \frac{d\Psi}{dt} \right)$$

$$e^{i\lambda} \left(-\hbar \frac{d\lambda}{dt} \Psi + i\hbar \frac{d\Psi}{dt} \right)$$

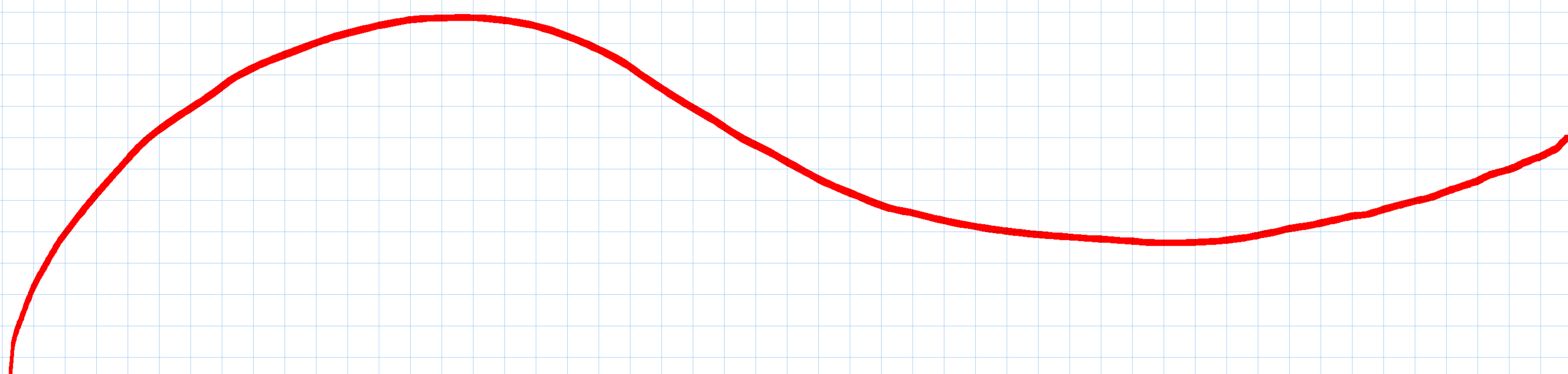
$$\cancel{e^{i\lambda}} \frac{1}{2\mu} \left(\frac{\hbar}{i} \nabla + \frac{e}{c} A \right) \left(\frac{\hbar}{i} \nabla + \frac{e}{c} (A + \hbar \nabla \lambda) \right) \Psi = \cancel{e^{i\lambda}} \left(-\hbar \frac{d\lambda}{dt} \Psi + i\hbar \frac{d\Psi}{dt} \right)$$

$$\Rightarrow \frac{1}{2\mu} \left(\frac{\hbar}{i} \nabla + \frac{e}{c} A \right) \left(\frac{\hbar}{i} \nabla + \frac{e}{c} (A + \hbar \nabla \lambda) \right) \Psi + \hbar \frac{d\lambda}{dt} \Psi - e\phi = i\hbar \frac{d\Psi}{dt}$$

$$\Rightarrow \frac{1}{2\mu} \left(\frac{\hbar}{i} \nabla + \frac{e}{c} A \right) \left(\frac{\hbar}{i} \nabla + \frac{e}{c} (A + \hbar \nabla \lambda) \right) \Psi - e \left(\phi - \frac{\hbar}{e} \frac{d\lambda}{dt} \right) \Psi = i\hbar \frac{d\Psi}{dt}$$

$$A' = A - \nabla f$$

$$\nabla f = \frac{\hbar c}{e} \nabla \lambda \Rightarrow \nabla \lambda = \frac{e}{\hbar c} \nabla f \Rightarrow \lambda = \frac{e}{\hbar c} f$$



Bir elektromagn. alandaki elektron için Sch Dnrk

$$\left[\frac{1}{2\mu} \left(\frac{\hbar}{i} \nabla + \frac{e}{c} A \right)^2 - e\phi \right] \Psi = i\hbar \frac{\partial \Psi}{\partial t} \quad \Psi = e^{-iEt/\hbar} \psi$$

$$\phi = 0 \quad B \neq 0$$

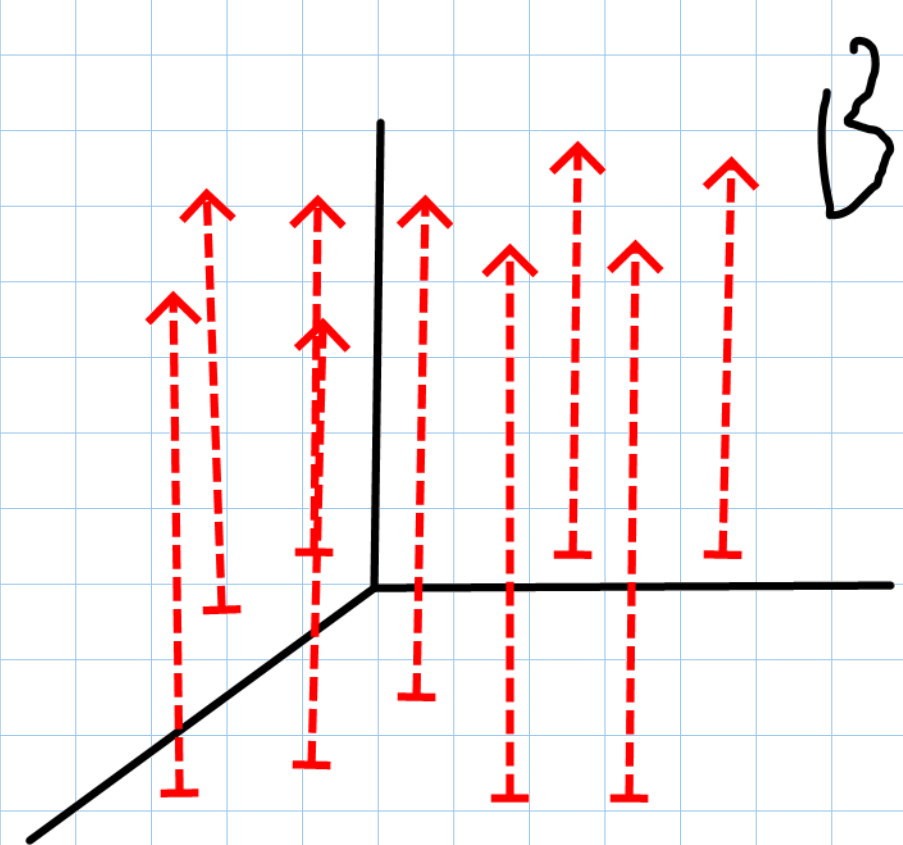
$$\frac{1}{2\mu} \left(\frac{\hbar}{i} \nabla + \frac{e}{c} A \right) \left(\frac{\hbar}{i} \nabla + \frac{e}{c} A \right) \psi = E \psi$$

$$\Rightarrow \left(\frac{\hbar}{i} \nabla + \frac{e}{c} A \right) \left(\frac{\hbar}{i} \nabla + \frac{e}{c} A \right) \psi = \left(\frac{\hbar}{i} \nabla + \frac{e}{c} A \right) \left(\frac{\hbar}{i} \nabla \psi + \frac{e}{c} A \psi \right) = -\hbar^2 \nabla^2 \psi + \hbar \frac{e}{i} \nabla A \psi + \frac{\hbar e}{i} A \nabla \psi + \frac{e^2}{c^2} A^2 \psi =$$

$$\nabla A = 0 //$$

$$-\hbar^2 \nabla^2 \psi + 2\hbar \frac{e}{i} A \nabla \psi + \frac{e^2}{c^2} A^2 \psi$$

$$\frac{1}{2\mu} \left(-\hbar^2 \nabla^2 \psi + 2\hbar \frac{e}{i} A \nabla \psi + \frac{e^2}{c^2} A^2 \psi \right) = E \psi \Rightarrow -\frac{\hbar^2}{2\mu} \nabla^2 \psi + \frac{\hbar e}{i\mu c} A \nabla \psi + \frac{e^2}{2\mu c^2} A^2 \psi = E \psi$$



$$B = B \hat{z}$$

$$A = -\frac{1}{2} r \times B$$

$$\nabla \times A = B$$

$$B_x \quad B_y \quad B_z$$

$$\vec{A} = -\frac{1}{2} (x\hat{x} + y\hat{y} + z\hat{z}) \times (B_x\hat{x} + B_y\hat{y} + B_z\hat{z})$$

$$\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ x & y & z \\ B_x & B_y & B_z \end{vmatrix}$$

$$\hat{i}(yB_z - zB_y) - \hat{j}(xB_z - zB_x) + \hat{k}(xB_y - yB_x)$$

$$B_z = B //$$

$$\Rightarrow \hat{i}(B_z(y - z)) - \hat{j}(xB_z - zB_x)$$

$$\Rightarrow B_z \{ \hat{x}(y - z) - \hat{y}(x - z) \}$$

$$A = -\frac{B}{2} (\hat{x}(y) - \hat{y}(x))$$

$$A = -\frac{B}{2} (y, x, 0)$$

Buna coulomb azeri der

$$A' = B(0, x, 0) \leftarrow \text{Landau azeri}$$

$$A' = A - \nabla f$$

$$\Rightarrow B(0, x, 0) = -\frac{B}{2} (y, x, 0) + \frac{B}{2} (y, \frac{B}{2}x, 0)$$

$$-\frac{\hbar^2}{2\mu} \nabla^2 \psi + \frac{\hbar}{\mu c} \frac{e}{c} A \nabla \psi + \frac{e^2}{2\mu c^2} A^2 \psi = E \psi$$

$$-\frac{\hbar^2}{2\mu} \nabla^2 \psi - \underbrace{i \frac{\hbar}{\mu c} \frac{e}{c} A \nabla \psi}_{\#4} + \underbrace{\frac{e^2}{2\mu c^2} A^2 \psi}_{\#3} = E \psi$$

$$\#3 \quad \frac{e^2}{2\mu c^2} A^2 = \frac{e^2}{2\mu c^2} \left(+ \frac{1}{4} (r \times B)^2 \right) = \frac{e^2}{8\mu c^2} (r \times B_z)^2 \quad (r \times B)_z = y\hat{x} - x\hat{y} = B_z$$

$$\Rightarrow \frac{e^2}{8\mu c^2} (y\hat{x} - x\hat{y})^2 \quad (y\hat{x} - x\hat{y}) (y\hat{x} - x\hat{y}) = \hat{y}^2 + \hat{x}^2$$

$\hat{y}\hat{x} = 0$
 $\hat{x}\hat{y} = 0$
 $\hat{y}\hat{y} = \hat{x}\hat{x} = 1$

$$\#3 = \frac{e^2}{8\mu c^2} (x^2 + y^2) \psi$$

$$\#4 \text{ kin } \frac{i\hbar}{\mu c} \frac{e}{c} A \nabla \psi \Rightarrow \frac{-i\hbar}{2\mu c} \frac{e}{c} (r \times B) \nabla \psi$$

$$\frac{-i\hbar}{2\mu c} e (-B) (r \times \nabla \psi) =$$

$$(A \times B) \cdot C = -B(A \times C)$$

$$(r \times B) \cdot \nabla \psi = -B(r \times \nabla \psi)$$

$$\Rightarrow \frac{e}{c} \frac{1}{2\mu} (-B) (r \times \nabla \psi) = -\frac{e}{c} \frac{1}{2\mu} B (r \times \nabla \psi) \Rightarrow -\frac{e}{2\mu c} B L \psi \quad B = B_z$$

$$\#4 \Rightarrow -\frac{eB}{2\mu c} L_z \psi \quad \text{Zeeman splitting}$$

$$L_z \psi = m \hbar \psi$$

$$-\frac{\hbar^2}{2\mu} \nabla^2 \psi - \frac{eB}{2\mu c} L_z \psi + \frac{e^2}{8\mu c^2} (x^2 + y^2) \psi = E \psi$$

$$-\frac{\hbar^2}{2\mu} \nabla^2 \psi - \left(\frac{eB}{2\mu c} L_z \psi \right) + \frac{e^2}{8\mu c^2} (x^2 + y^2) \psi =$$

$$\frac{eB}{2\mu c} = \omega_L \quad \text{Larmor frequency}$$

$$-\frac{\hbar^2}{2\mu} \nabla^2 \psi + \omega_L L_z \psi + \frac{e^2}{8\mu c^2} (x^2 + y^2) \psi$$

$$x^2 + y^2 = \rho^2$$

$$-\frac{eB}{2\mu c} L_z \psi = -\frac{eB}{2\mu c} m \hbar \psi \Rightarrow -\frac{eB \hbar m}{2\mu c} \psi$$

$$\text{I} \dots (\#4) = \frac{eB \hbar}{2\mu c} \quad \# = \frac{e^2 B_0^2}{8\mu c} \rho^2$$

$$\text{II} \dots E_n = \frac{(\mu c)^2}{2} \frac{(2\alpha)^2}{n^2} \approx \frac{e^2}{2\alpha} \dots \text{III}$$

$$\frac{\text{II}}{\text{I}} \approx \frac{B}{9 \times 10^9 \text{ Gauss}} \quad \frac{\text{I}}{\text{III}} \approx \frac{B}{5 \times 10^9 \text{ Gauss}}$$

Perturbations sind